GWH DOCUMENTATION SOURCE SHEET

INTERNAL ORGANISATION

GWH gathers genomic assembly data that are directly submitted to GWH or integrated from NCBI (GenBank or RefSeq).

Data files can be retrieved from https://download.cncb.ac.cn/.
The accession paths vary depending on the data source.

Finding assemblies and associated data files

1. Directly submitted to GWH

The records follow the classical structure:

```
BioProject
PRJCA......
   |
   +-- BioSample
   |   SAMC......
   |       |
   |       +-- Assembly / Genome
   |       |   GWH.......
   |       |
   |       +-- Assembly / Genome
   |           GWH.......
   |
   +-- BioSample
       SAMC......
```

Data files can be retrieved at:
https://download.cncb.ac.cn/gwh/

The file organisation is as follows:

Direct submissions:

```
/gwh/
   Animals/
   Archaea/
```

Bacteria/
Fungi/
Metagenomes/
Others/
Plants/
Protists/
Viruses/

Within each organism category:

<Scientific_name>_<Assembly_name>_<GWH_accession>/
<GWH_accession>.genome.fasta.gz
<GWH_accession>.gff.gz
<GWH_accession>.RNA.fasta.gz
<GWH_accession>.CDS.fasta.gz
<GWH_accession>.Protein.faa.gz
<GWH_accession>.feature.gz
<GWH_accession>_md5.txt

Reannotations produced by GWH:

/gwh/Reannotation/
|
+-- Archaea/
| +-- <XX>/
| +-- GWHRA_<ACCESSION>.<version>/
|
+-- Bacteria/
+-- AA/
+-- AB/
+-- AC/
+-- ...
+-- DU/
+-- GWHRA_DUBW00000000.2/
GWHRA_DUBW00000000.2.genome.fasta.gz
GWHRA_DUBW00000000.2.gff.gz
GWHRA_DUBW00000000.2.RNA.fasta.gz
GWHRA_DUBW00000000.2.CDS.fasta.gz
GWHRA_DUBW00000000.2.Protein.faa.gz
GWHRA_DUBW00000000.2.feature.gz
*_md5.txt

2. Integrated from NCBI

Data files can be retrieved at:
https://download.cncb.ac.cn/assembly/ncbi/

Integrated NCBI assemblies mirrored by GWH:

/assembly/ncbi/
- Animals/
- Archaea/
- Bacteria/
- Fungi/
- Metagenomes/
- Others/
- Plants/
- Protists/
- Viruses/

Within each organism category:

GCA/ (assemblies from GenBank)
- <first three digits>/
  - <next three digits>/
    - <organism>_<assembly_name>_<GCA_accession.version>/
      - <GCA_accession.version>_<assembly_name>_genomic.fna.gz
      - <GCA_accession.version>_md5.txt

GCF/ (assemblies from RefSeq)
- <first three digits>/
  - <next three digits>/
    - <organism>_<assembly_name>_<GCF_accession.version>/
      - <GCF_accession.version>_<assembly_name>_genomic.fna.gz
      - ...

Example path:

Home / assembly / ncbi / Animals / GCA / 000 / 002 /
Apis_mellifera_honey_bee_Amel_4.5_GCA_000002195.1

The final directory contains the data associated with the assembly.
The minimum files identified are:

Accession_ID_genomic.fna.gz

Accession_ID_md5.txt

Accession prefixes help identify the source when parsing the records:

GWHxxxx...   : assembly submitted to GWH
GWHRA_xxxx... : reannotation generated by GWH
GCA_xxxxx... : GenBank assembly imported from NCBI
GCF_xxxxx... : RefSeq assembly imported from NCBI

Finding statistics files

Statistics are accessible from assembly webpages. Dedicated statistics files were not identified in the download directories inspected at https://download.cncb.ac.cn/.

The available statistics vary considerably between assemblies.

Example: Canis lupus familiaris, accession GWHBJIX01000000.

Statistics of Genome Assembly:
  Genome size (bp): 16,751
  GC content: 39.68%

Note: this sequence length should not be interpreted as the size of the complete nuclear genome of the species, even if the record is labelled "Full Genome".

Differences between NCBI/RefSeq and GWH

NCBI / RefSeq assembly directories may contain:
  FASTA
  GFF
  GTF
  assembly_report
  assembly_stats
  annotation report
  feature tables
  evidence
  additional pipeline artefacts

Direct GWH assembly directories generally have a simpler structure:
  genome FASTA
  GFF

RNA
CDS
proteins
features
MD5

File availability depends on the assembly and its annotation.
Statistics are mainly displayed in the database or on the webpage.
Quality control is primarily performed during the submission pipeline.

INTERNAL PIPELINES

Submission pipeline in GWH

1. GSA reads or reads from another source.

2. Assembly performed by the submitter.

3. Preparation of the submission:
   - FASTA assembly
   - Metadata
   - GFF/TBL annotation files (optional)

4. Association with:
   - BioProject PRJCA...
   - BioSample SAMC...

5. Submission to GWH:
   - Automated checks
   - Corrections by the submitter if errors are detected

6. Assignment of a GWH accession:
   - GWHxxxx00000000[.version]

7. Release.

8. For prokaryotic genomes, GWH provides an automated reannotation pipeline based on NCBI PGAP. Reannotation results receive a separate accession.

Updates to assemblies and annotations

For an initial GWH submission, the base accession follows this model:

GWHAAEV00000000

Updates are distinguished by appending ".1", then ".2", and so on:

accession_core = GWHAAEV00000000
version = 2
accession = GWHAAEV00000000.2

Reannotation accessions may also include a version suffix:

GWHRA_DUBW00000000
GWHRA_DUBW00000000.1

The full accession, including any version suffix, should be preserved.